

Introduction to prosthodontic Dentistry

prosthodontics

- ▶ Branch of dentistry pertaining to the maintenance of oral function by the replacement of missing teeth and associated structures by artificial devices(prosthesis)
- ▶ **Dental Prosthesis:** An artificial replacement of missing teeth and associated structures(e.g. crowns, dentures, bridges...)
- ▶ When there are missing teeth, we put artificial teeth in their place.



Definitions

- ▶ **Dentulous/dentate:** An individual who has his/her natural teeth present.



- ▶ **Edentulous:** The individual who has lost his/her natural teeth.
- ▶ Depending on the number of teeth missing they may be partially or completely edentulous.
- ▶ Partially edentulous: one or more of the natural teeth is missing but not all of the natural teeth are missing.
- ▶ Completely edentulous: all of the natural teeth are missing in upper, lower or both jaws.



Reasons for tooth loss

- ▶ Traumatic injuries
- ▶ Caries
- ▶ Periodontal diseases
- ▶ Cysts, malignancies and tumors
- ▶ Iatrogenic extraction
- ▶ Congenitally missing teeth
- ▶ Failure to erupt (impacted teeth)



Anatomical consequences of tooth loss



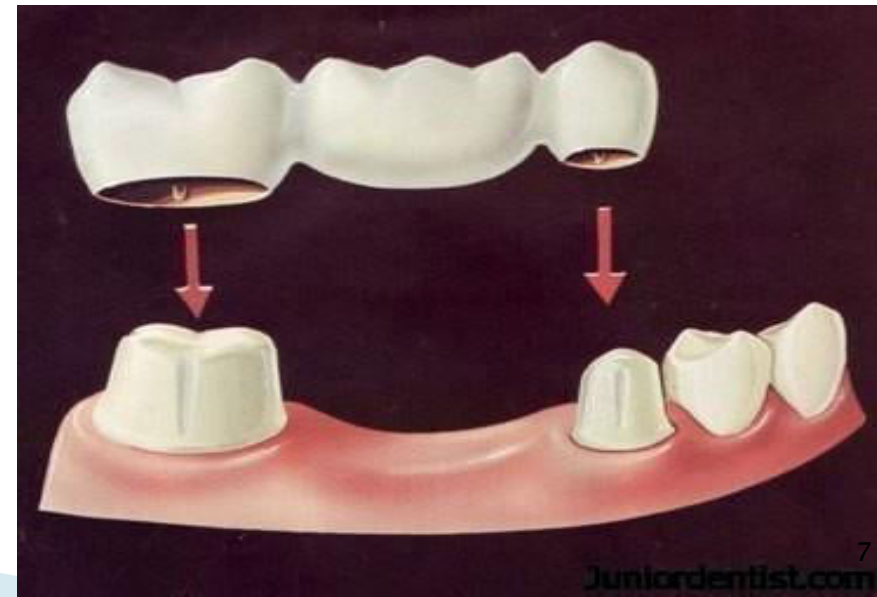
Branches of Prosthodontics

1. Fixed prosthodontics:

A- crown



B- bridge



2. Removable prosthodontics

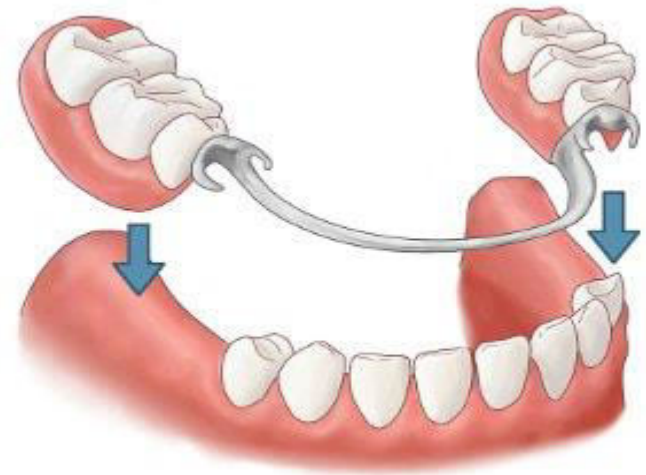
a) complete denture:

A dental prosthesis which replaces the entire dentition and associated structures of the maxilla and mandible”



B) Removable Partial denture

Is a prosthesis that replaces some teeth in a partially dentate arch, and can be removed from the mouth and replaced



Difference between FPD and RPD

- ▶ **Partial denture** can be described as a **fixed dental prosthesis** or **removable dental prosthesis**. Based on the patient's capability to remove and put back the prosthesis by himself in his mouth.
- ▶ The **fixed dental prosthesis** usually can only be removed by the dentist.

3– Maxillofacial prosthodontics



Nasal prosthesis

Auricular
prosthesis



4. Implant prosthodontics

- ▶ Concerning the replacement of missing teeth and/or associated structures by restorations that are attached to dental implants.
- ▶ Dental Implant: artificial root that replaces the natural tooth root. which is inserted within the jaw bone to provide retention and support for fixed or removable dental prosthesis.



Fixed Prosthodontics

- ▶ **Crown(cap):**
- ▶ when large portion of crown has been destroyed by dental caries & has had filling is broken or worn, the tooth must be covered by artificial crown.

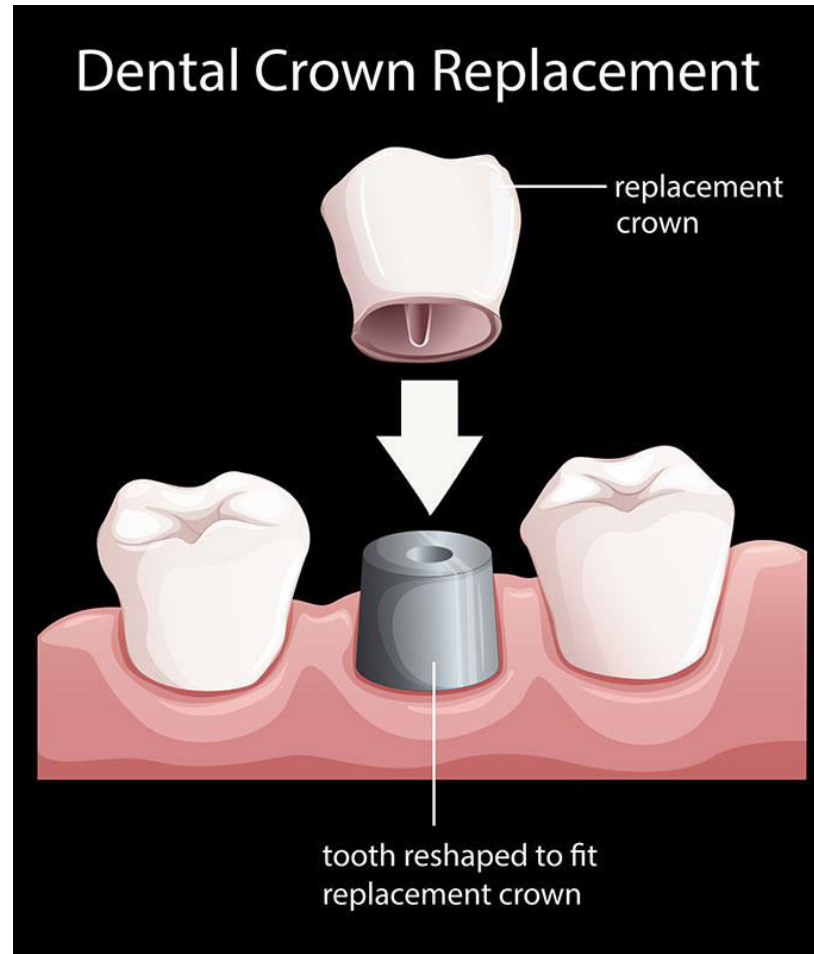


- ▶ Crowns are cemented into place, and they replace enamel & some of the dentin.

Single Crown



The single crown replacement



Fixed Prosthodontics

Dental Bridge(Fixed partial denture)

- ▶ If a single tooth or few teeth are missing, the enamel may be removed from teeth adjacent(neighboring) to the space & fixed prosthesis (bridge) may be cemented into place to fill the space created by tooth removal.
- ▶ Crowns and bridges are considered as fixed prosthodontics(cemented to the remaining natural teeth)





Missing tooth



Adjacent teeth prepared



Bridge created



Bridge in place

- ▶ Bridges(fixed) are cemented into place, and they replace missing tooth or several teeth.





Pontic: artificial tooth that replaces a missing tooth, restores its functions and fills the space.

Retainer: takes support from abutment tooth and provide retention for the prosthesis.

Abutment: the prepared tooth supporting the bridge adjacent to the missing teeth (space).

Fixed Prosthodontics

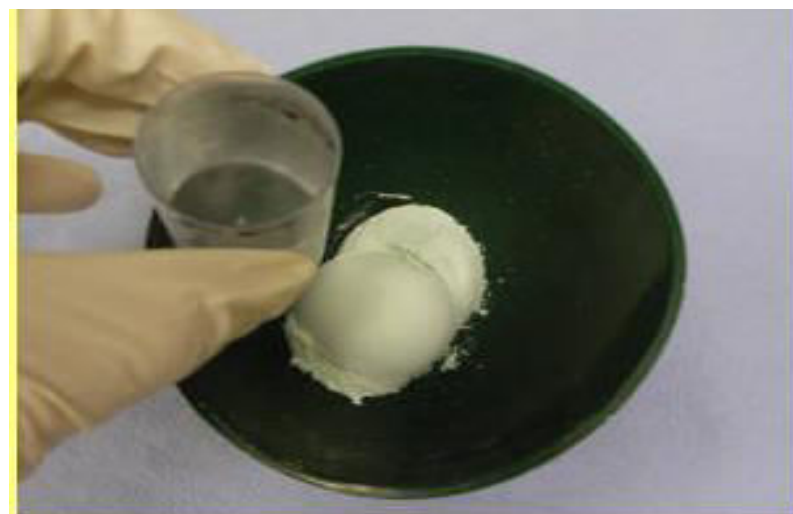
- ▶ Crowns & bridges could be made from metal alone, metal with tooth colored porcelain, or ceramic alone



Impression

- ▶ Primary impression is the **first step** in fabrication of any prosthesis , to make a **copy or print** of the oral structures that will receive prosthesis, usually made by **alginate** material.





Tray: A device used to carry, confine and control the impression material while making an impression



Impression is poured by gypsum to fabricate a model (cast) that **mimic** the oral cavity structures

Cast: Is a positive likeness (copy) of the denture bearing tissues



The cast



Fixed prosthodontics

- ▶ Fixed prosthodontics is one of the most accepted, long term & potentially pleasing in appearance.
- ▶ The high cost is a major drawback.

Costs are high because:

- ▶ Clinical skills necessary
- ▶ Expenses for significant laboratory time
- ▶ Cost of materials

Fixed prosthodontics procedures

- ▶ Dentists **prepare** the teeth by trimming, then take impression then send them to the **lab**, the laboratory technicians construct the crowns or bridges then send them back to the dentist who seats them & cement them.

Teeth with large restorations and unpleasant appearance



The teeth preparation



The teeth preparation



The teeth are prepared



Taking the impression





The Final Restoration(Bridge)





Fixed prosthodontics

- ▶ **Clinical situation** where fixed prosthodontics may interfere:
 - 1–One missing tooth
 - 2–Missing teeth
 - 3–Broken tooth
 - 4–Teeth with large dental caries ➡ (Large fillings)
 - 5–Teeth with defective filling

Treatment options for the broken tooth, large caries, defective filling

- ▶ 1–Crown(cap) metal, porcelain,...etc
- ▶ 2–Tooth colored filling(e.g. composite)
- ▶ 3–Metallic filling (amalgam)
- ▶ 4–Extraction & placement of fixed or removable prosthesis
- ▶ 5–Extraction & placement of implant



Implant



4 Options to Replace a Tooth



Types of Bridges

- ▶ **1 –Traditional fixed prosthesis(bridge):** at least one tooth on each side of space.



- ▶ **2-Maryland bridge:** where **less** prepared (trimmed) teeth on each side of the space are used, with a bridge holding a false tooth and usually leaving the **front portion** of supporting teeth intact. (two neighboring teeth lingually prepared)



- ▶ **3-Cantilever fixed prosthesis:** traditional bridge involving one or more teeth on one side of space only.



Removable prosthodontics

- ▶ **complete denture** are usually made of acrylic resin material (plastic like material) .

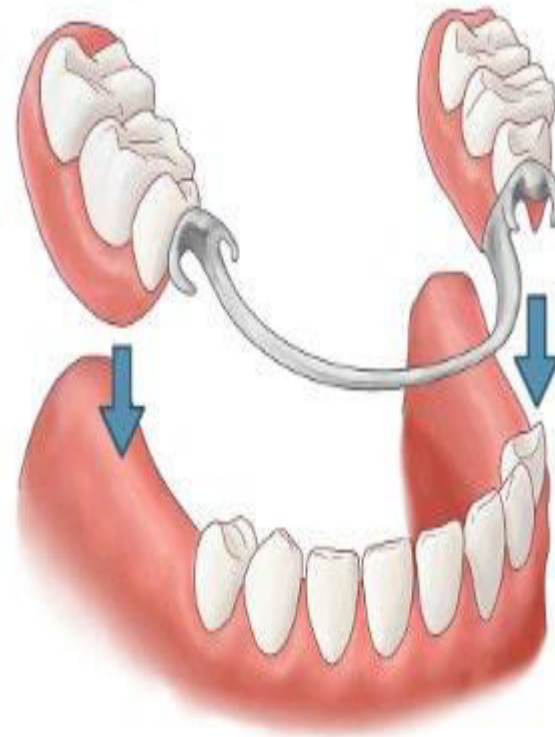


Partial denture(RPD)

- ▶ Partial denture made of **acrylic resin** or **acrylic resin with metals** like chrome cobalt, or nickel



Removable partial dentures



Removable prosthodontics

- ▶ Removable acrylic dentures cost is generally low, but metallic partial dentures are relatively expensive.
- ▶ Removable dentures may need frequent repair & modifications.

Removable prosthodontics

- ▶ Removable dentures need a continuous maintenance & care
- ▶ Cleaning the dentures with special brushes & disinfectant agents.(as sodium hypochlorite and chlorhexidine)
- ▶ The denture must be clean from any food accumulation or calcifications.

Splinting



THANK YOU